Contribution Title (How to call it?)

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Abstract. **Abstract.** Climate science relies on large raster datasets that are rarely accessible to non-expert users. Existing tools either require GIS expertise or offer only static visualisations with no ability to compare locations or time periods interactively.

This paper presents [SystemName], a web application that provides interactive access to WorldClim historical climate data with the ability to retrieve climate statistics for a specified location but also to compare cities and compare climate for different time periods. Finally user also can render heatmap on the map for selected location.

Key capabilities include both: interactive standard bars and Walter-Lieth climatograms with conventional arid and humid period shading, a spatially resolved regional heatmap rendered directly from WorldClim raster pixels, and plots export functionality.

Application is ready to use without any no registration or local software installation. It is localized in 10 languages with fast multilingual city search.

In addition, the evaluation of the application was performed with 27 European students, whose level in climatology was different. The System Usability Scale (SUS) score obtained 84.6 points which is close to excellent. The conclusion is that [SystemName] is user-friendly app which makes it easy for everyone to visualize and explore climate data.

Keywords: Climate Data Visualisation · WorldClim · GIS · Walter-Lieth Climatogram · Usability Evaluation

1 Introduction

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2 Related Work

2.1 WorldClim

2.2 Existing Climate Visualisation Tools

3 Web Application

3.1 Requirements

3.2 Data Sources and Integration

3.3 Architecture

3.4 Main Features

4 Evaluation

4.1 Scenarios

4.2 Participants and Procedure

4.3 Results and Feedback

4.4 Discussion and Limitations

5 Conclusions